

GRNsight v7: a web application for visualizing gene regulatory networks and protein-protein interactions

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GRNsight is an open-source web application for visualizing models of gene regulatory networks (GRNs) and protein-protein physical interaction networks (PPIs), either uploaded by the user or obtained from our backend database. Graph models display rectangular nodes, representing genes/proteins, with edges connecting them. GRNs have directed edges to indicate regulatory relationships (activation or repression), with edge thickness indicating regulation strength. Biologists can then compare the edges to timecourse expression data displayed on the nodes (either user-uploaded or from our database) to validate the GRN model. PPIs have undirected edges to indicate physical binding. For v7, we improved our PPI support, transitioned our database loading scripts from YeastMine to AllianceMine (as YeastMine was discontinued), and enhanced viewport functionality. Users can now color nodes in PPIs using expression data. The top and bottom of nodes can display different datasets. New graph functionality allows users to restrict the graph to the viewport. Users can drag, zoom, and move the graph with nodes staying within the application's bounding box, improving user control of the graph. Finally, the project's CI pipeline has migrated from Travis CI to GitHub Actions, integrating workflows for automated builds, unit tests, and linting to streamline development processes and improve maintainability of the code. Ongoing efforts prioritize bug resolution, user interface enhancements, and improved documentation. GRNsight is now positioned to comprehensively address GRNs and PPIs, offering a unified platform for visualizing diverse molecular interaction models. GRNsight is available at <https://dondi.github.io/GRNsight/>.